

RLHF Objective

- How do we actually change our LM parameters θ to maximize this?

$$\mathbb{E}_{\hat{s} \sim p_{\theta}(s)}[R(\hat{s})]$$

- Let's try doing gradient ascent!

$$\theta_{t+1} := \theta_t + \alpha \nabla_{\theta_t} \mathbb{E}_{\hat{s} \sim p_{\theta_t}(s)}[R(\hat{s})]$$

How do we estimate
this expectation??



What if our reward
function is non-
differentiable??



$$\nabla_{\theta} \sum_s R(s) \underbrace{p_{\theta}(s)} = \sum_s R(s) \underbrace{\nabla_{\theta} p_{\theta}(s)}$$

$$\nabla_{\theta} \log p_{\theta}(s) = \frac{1}{p_{\theta}(s)} \nabla_{\theta} p_{\theta}(s)$$

$$= \sum_s p_{\theta}(s) R(s) \nabla_{\theta} \log p_{\theta}(s)$$

$$= \mathbb{E}_{\hat{s} \sim p_{\theta}(s)} \left[R(\hat{s}) \nabla_{\theta} \log p_{\theta}(\hat{s}) \right]$$

$$\approx \frac{1}{n} \sum_{i=1}^n R(s_i) \nabla_{\theta} \log p_{\theta}(s_i)$$

$$Q_{t+1} = Q_t + \alpha R(s) \nabla_{\theta_t} \log p_{\theta_t}(s) \quad \rightarrow \text{not GT}$$

$$- \alpha \nabla_{\theta_t} \log p_{\theta_t}(s)$$

Using Policy Gradient

- We want to obtain

$$\nabla_{\theta} \mathbb{E}_{\hat{s} \sim p_{\theta}(s)}[R(\hat{s})] = \nabla_{\theta} \sum_s R(s) p_{\theta}(s) = \sum_s R(s) \nabla_{\theta} p_{\theta}(s)$$

(defn. of expectation) (linearity of gradient)

- Here we'll use a very handy trick known as the **log-derivative trick**. Let's try taking the gradient of $\log p_{\theta}(s)$

$$\nabla_{\theta} \log p_{\theta}(s) = \frac{1}{p_{\theta}(s)} \nabla_{\theta} p_{\theta}(s) \quad \Rightarrow \quad \nabla_{\theta} p_{\theta}(s) = p_{\theta}(s) \nabla_{\theta} \log p_{\theta}(s)$$

(chain rule)

- Plug back in:

$$\sum_s R(s) \nabla_{\theta} p_{\theta}(s) = \sum_s p_{\theta}(s) R(s) \nabla_{\theta} \log p_{\theta}(s)$$

This is an expectation of this

$$= \mathbb{E}_{\hat{s} \sim p_{\theta}(s)}[R(\hat{s}) \nabla_{\theta} \log p_{\theta}(\hat{s})]$$

Using Policy Gradient

- Now we have put the gradient “inside” the expectation, we can approximate this objective with Monte Carlo samples:

$$\nabla_{\theta} \mathbb{E}_{\hat{s} \sim p_{\theta}(s)}[R(\hat{s})] = \mathbb{E}_{\hat{s} \sim p_{\theta}(s)}[R(\hat{s}) \nabla_{\theta} \log p_{\theta}(\hat{s})] \approx \frac{1}{m} \sum_{i=1}^m R(s_i) \nabla_{\theta} \log p_{\theta}(s_i)$$

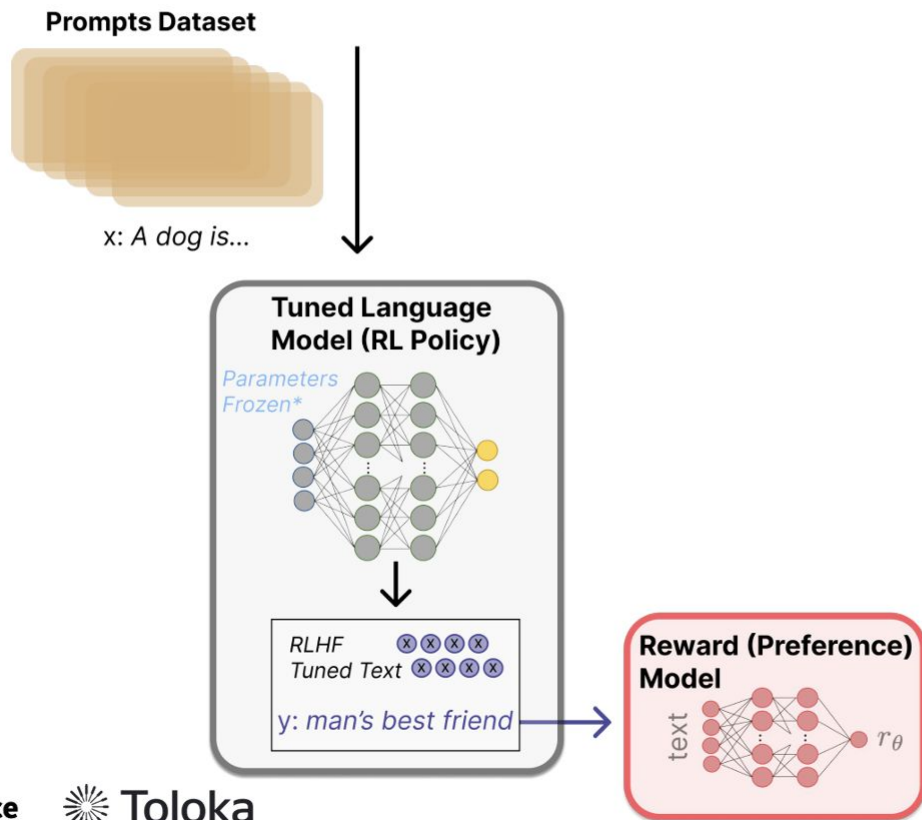
- Giving us the update rule:

$$\theta_{t+1} := \theta_t + \alpha \frac{1}{m} \sum_{i=1}^m R(s_i) \nabla_{\theta_t} \log p_{\theta_t}(s_i)$$

Annotations for the update rule:

- If R is +++** (Green arrow pointing to $R(s_i)$)
- Take gradient steps to maximize $p_{\theta}(s_i)$** (Green arrow pointing to $\nabla_{\theta_t} \log p_{\theta_t}(s_i)$)
- If R is ---** (Red arrow pointing to $R(s_i)$)
- Take steps to minimize $p_{\theta}(s_i)$** (Red arrow pointing to $\nabla_{\theta_t} \log p_{\theta_t}(s_i)$)
- Not GT model completion** (Handwritten note with an arrow pointing to the gradient term)

Fine tuning with RL - using a reward model

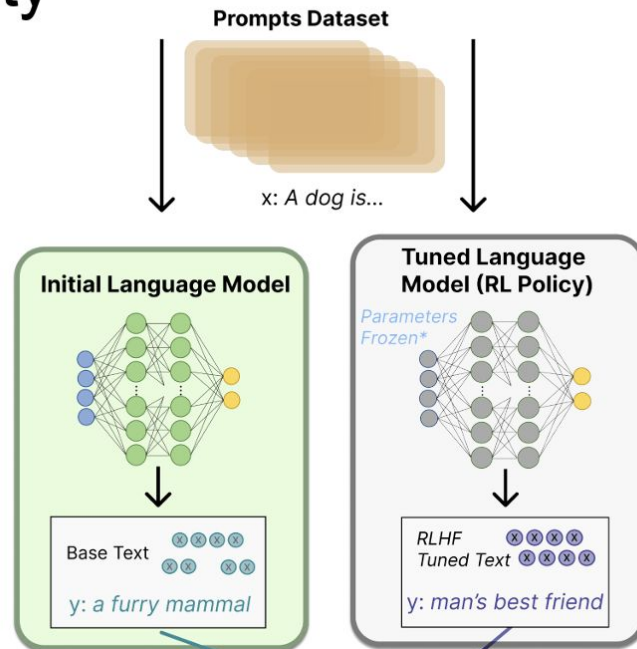


Fine tuning with RL - KL penalty

Kullback–Leibler (KL) divergence: $D_{\text{KL}}(P \parallel Q)$

Distance between distributions

Constrains the RL fine-tuning to not result in a LM that outputs gibberish (to fool the reward model).



Handwritten note: *Hyperparameters*

$$-\lambda_{\text{KL}} D_{\text{KL}}(\pi_{\text{PPO}}(y|x) \parallel \pi_{\text{base}}(y|x))$$

Handwritten notes: *KL prediction shift penalty* (under $\pi_{\text{PPO}}(y|x)$) and *RLHF* (under $\pi_{\text{base}}(y|x)$)

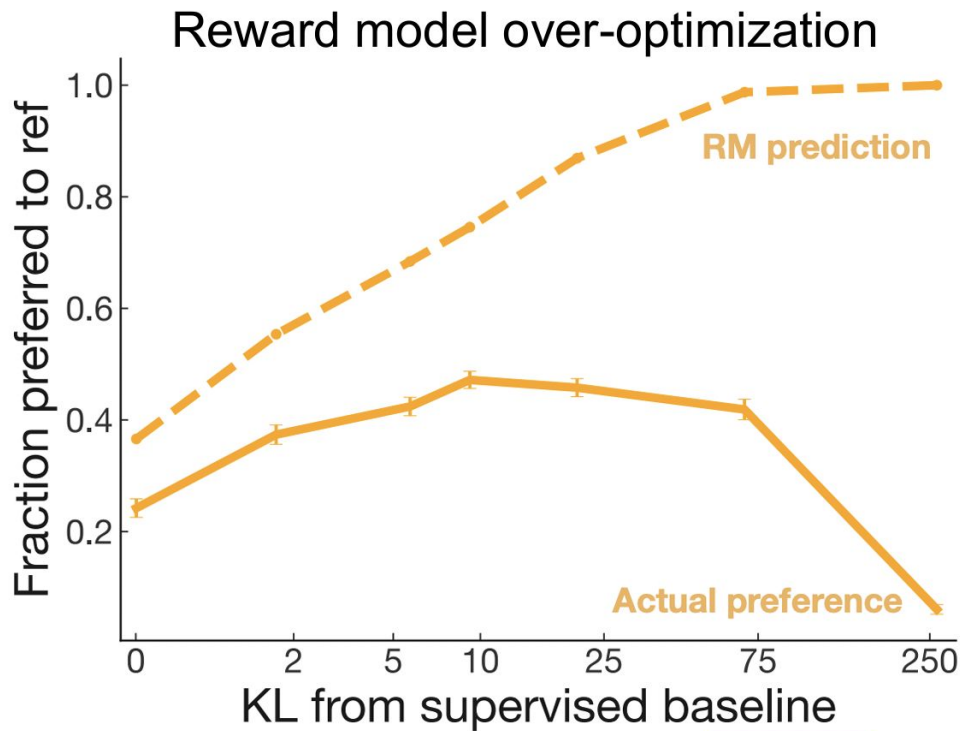


Hugging Face



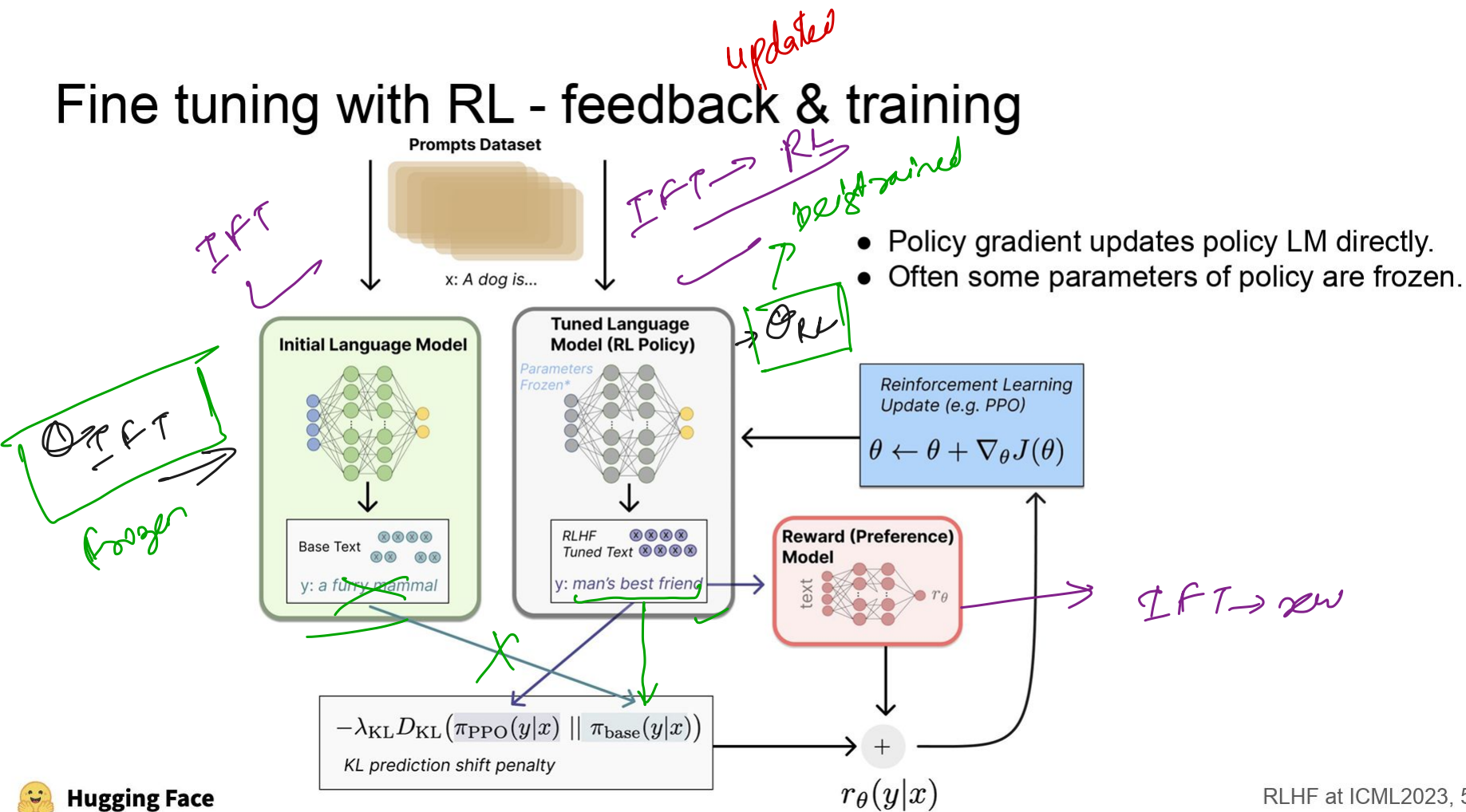
Toloka

RLHF: use of KL divergence



$$R(s) = \underbrace{RM_{\phi}(s)} - \underbrace{\beta \log \left(\frac{p_{\theta}^{RL}(s)}{p^{PT}(s)} \right)} \quad \rightarrow$$

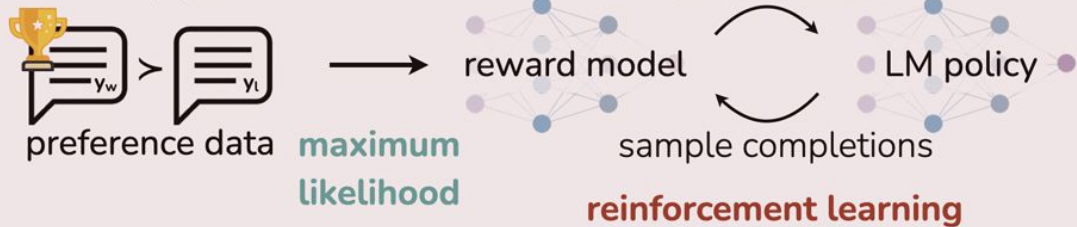
Fine tuning with RL - feedback & training



Direct Preference Optimization (DPO)

Reinforcement Learning from Human Feedback (RLHF)

x: "write me a poem about
the history of jazz"



Direct Preference Optimization (DPO)

x: "write me a poem about
the history of jazz"

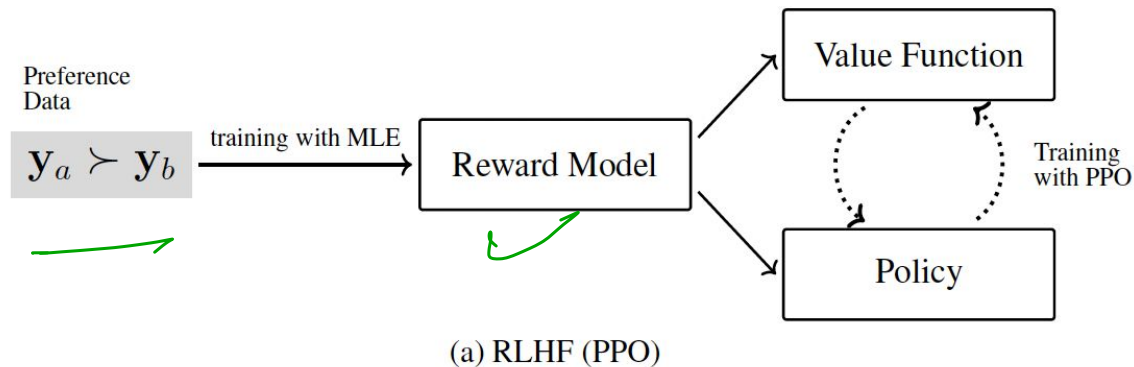


Training a reliable reward model is itself not an easy task, and a poorly trained reward model can greatly affect the outcome of policy learning.

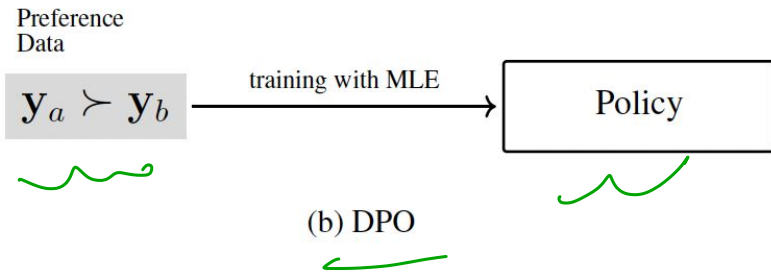
- Direct preference optimization (DPO) simplifies the training framework by eliminating the need to explicitly model rewards
- It directly optimizes the policy based on user preferences, rather than developing a separate reward model

Direct Preference Optimization: Your Language Model is Secretly a Reward Model

Direct Preference Optimization (DPO)

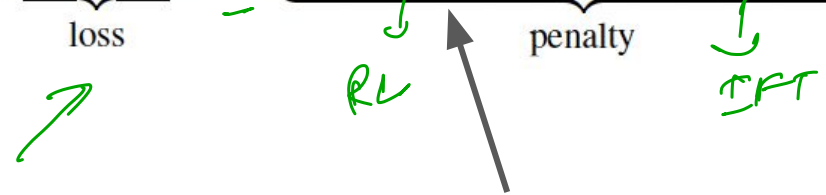


In DPO, the use of human preference data is more direct, and the policy is trained on this data without the need for reward model training.



DPO Objective

Let's start from the (simple) RLHF loss function where the quality of the output y given the input x is evaluated by the reward model $r(x, y)$, and a KL divergence term

$$\tilde{\theta} = \arg \min_{\theta} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} \mathbb{E}_{\mathbf{y} \sim \pi_{\theta}(\cdot|\mathbf{x})} \left[\underbrace{-r(\mathbf{x}, \mathbf{y})}_{\text{loss}} + \beta \underbrace{(\log \pi_{\theta}(\mathbf{y}|\mathbf{x}) - \log \pi_{\theta_{\text{ref}}}(\mathbf{y}|\mathbf{x}))}_{\text{penalty}} \right]$$


Only this term depends
on the target policy

$$\min \left[\log \pi_\theta(y|x) - \left(\log \pi_{\theta_{\text{ref}}}(y|x) + \frac{1}{\beta} r(x,y) \right) \right]$$

$$= \min \left[\log \pi_\theta(y|x) - \log \left(\pi_{\theta_{\text{ref}}}(y|x) e^{\frac{1}{\beta} r(x,y)} \right) \right]$$

$Z(x) = \sum_y \left[\pi_{\theta_{\text{ref}}}(y|x) e^{\frac{1}{\beta} r(x,y)} \right]$ should have been

$\log \pi^*(y|x)$

$$- \log \left[\frac{\pi_{\theta_{\text{ref}}}(y|x) e^{\frac{1}{\beta} r(x,y)}}{Z(x)} \times Z(x) \right]$$

$$= \min \left[\log \pi_\theta(y|x) - \log \pi^*(y|x) - \log Z(x) \right]$$

DPO Objective

$$\tilde{\theta} = \arg \min_{\theta} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} \mathbb{E}_{\mathbf{y} \sim \pi_{\theta}(\cdot|\mathbf{x})} \left[\underbrace{-r(\mathbf{x}, \mathbf{y})}_{\text{loss}} + \underbrace{\beta (\log \pi_{\theta}(\mathbf{y}|\mathbf{x}) - \log \pi_{\theta_{\text{ref}}}(\mathbf{y}|\mathbf{x}))}_{\text{penalty}} \right]$$

Depends on policy
↓

we rearrange the right-hand side of Eq. to isolate this as an independent term:

$$\begin{aligned} \tilde{\theta} &= \arg \min_{\theta} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} \mathbb{E}_{\mathbf{y} \sim \pi_{\theta}(\cdot|\mathbf{x})} [\beta \log \pi_{\theta}(\mathbf{y}|\mathbf{x}) - \beta \log \pi_{\theta_{\text{ref}}}(\mathbf{y}|\mathbf{x}) - r(\mathbf{x}, \mathbf{y})] \\ &= \arg \min_{\theta} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} \mathbb{E}_{\mathbf{y} \sim \pi_{\theta}(\cdot|\mathbf{x})} [\log \pi_{\theta}(\mathbf{y}|\mathbf{x}) - (\log \pi_{\theta_{\text{ref}}}(\mathbf{y}|\mathbf{x}) + \frac{1}{\beta} r(\mathbf{x}, \mathbf{y}))] \\ &= \arg \min_{\theta} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} \mathbb{E}_{\mathbf{y} \sim \pi_{\theta}(\cdot|\mathbf{x})} [\underbrace{\log \pi_{\theta}(\mathbf{y}|\mathbf{x})}_{\text{dependent on } \theta} - \underbrace{\log \pi_{\theta_{\text{ref}}}(\mathbf{y}|\mathbf{x}) \exp(\frac{1}{\beta} r(\mathbf{x}, \mathbf{y}))}_{\text{not dependent on } \theta}] \end{aligned}$$

DPO Objective

$$\tilde{\theta} = \arg \min_{\theta} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} \mathbb{E}_{\mathbf{y} \sim \pi_{\theta}(\cdot|\mathbf{x})} \left[\underbrace{\log \pi_{\theta}(\mathbf{y}|\mathbf{x})}_{\text{dependent on } \theta} - \underbrace{\log \pi_{\theta_{\text{ref}}}(\mathbf{y}|\mathbf{x}) \exp\left(\frac{1}{\beta} r(\mathbf{x}, \mathbf{y})\right)}_{\text{not dependent on } \theta} \right]$$

This equation defines the objective function as the difference between the log-probability distribution function of $\mathbf{y}$ and another function of $\mathbf{y}$.

This form of the objective function seems not “ideal”, as we usually prefer to see the difference between two distributions, so that we can interpret this difference as some kind of divergence between the distributions.

A simple idea is to convert the second term into a log-probability distribution over the domain of $\mathbf{y}$

DPO Objective

$$\tilde{\theta} = \arg \min_{\theta} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} \mathbb{E}_{\mathbf{y} \sim \pi_{\theta}(\cdot|\mathbf{x})} \left[\underbrace{\log \pi_{\theta}(\mathbf{y}|\mathbf{x})}_{\text{dependent on } \theta} - \underbrace{\log \pi_{\theta_{\text{ref}}}(\mathbf{y}|\mathbf{x}) \exp\left(\frac{1}{\beta} r(\mathbf{x}, \mathbf{y})\right)}_{\text{not dependent on } \theta} \right]$$

See the second term as log of an unnormalized probability of $\mathbf{y}$, we can convert it into a normalized probability by dividing it by a normalization factor.

$$Z(\mathbf{x}) = \sum_{\mathbf{y}} \pi_{\theta_{\text{ref}}}(\mathbf{y}|\mathbf{x}) \exp\left(\frac{1}{\beta} r(\mathbf{x}, \mathbf{y})\right)$$

Hence we can define a probability distribution by

$$\pi^*(\mathbf{y}|\mathbf{x}) = \frac{\pi_{\theta_{\text{ref}}}(\mathbf{y}|\mathbf{x}) \exp\left(\frac{1}{\beta} r(\mathbf{x}, \mathbf{y})\right)}{Z(\mathbf{x})}$$

DPO Objective

$$\tilde{\theta} = \arg \min_{\theta} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} \mathbb{E}_{\mathbf{y} \sim \pi_{\theta}(\cdot|\mathbf{x})} \left[\underbrace{\log \pi_{\theta}(\mathbf{y}|\mathbf{x})}_{\text{dependent on } \theta} - \underbrace{\log \pi_{\theta_{\text{ref}}}(\mathbf{y}|\mathbf{x}) \exp\left(\frac{1}{\beta} r(\mathbf{x}, \mathbf{y})\right)}_{\text{not dependent on } \theta} \right]$$

$$= \arg \min_{\theta} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} \mathbb{E}_{\mathbf{y} \sim \pi_{\theta}(\cdot|\mathbf{x})} \left[\log \pi_{\theta}(\mathbf{y}|\mathbf{x}) - \log \frac{\pi_{\theta_{\text{ref}}}(\mathbf{y}|\mathbf{x}) \exp\left(\frac{1}{\beta} r(\mathbf{x}, \mathbf{y})\right)}{Z(\mathbf{x})} - \log Z(\mathbf{x}) \right]$$

*Rewriting
through $Z(\mathbf{x})$*

$$= \arg \min_{\theta} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} \mathbb{E}_{\mathbf{y} \sim \pi_{\theta}(\cdot|\mathbf{x})} \left[\log \pi_{\theta}(\mathbf{y}|\mathbf{x}) - \log \pi^*(\mathbf{y}|\mathbf{x}) - \log Z(\mathbf{x}) \right] \quad \checkmark$$

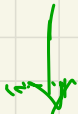
$$= \arg \min_{\theta} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} \left[\mathbb{E}_{\mathbf{y} \sim \pi_{\theta}(\cdot|\mathbf{x})} \left[\log \pi_{\theta}(\mathbf{y}|\mathbf{x}) - \log \pi^*(\mathbf{y}|\mathbf{x}) \right] - \mathbb{E}_{\mathbf{y} \sim \pi_{\theta}(\cdot|\mathbf{x})} [\log Z(\mathbf{x})] \right]$$

$$= \arg \min_{\theta} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} \left[\underbrace{\text{KL}(\pi_{\theta}(\cdot|\mathbf{x}) \parallel \pi^*(\cdot|\mathbf{x}))}_{\text{KL divergence}} - \underbrace{\log Z(\mathbf{x})}_{\text{constant wrt. } \theta} \right] \quad \checkmark$$

minimize my loss when

$$\pi_{\theta}(\cdot|x) = \pi^*(\cdot|x)$$

$$\Rightarrow \frac{\pi_{\theta_{\text{ref}}}(y|x) \exp\left(\frac{1}{\beta} r(x,y)\right)}{Z(x)} = \pi_{\theta}(y|x)$$

 rearrangement

$$r(x,y) = \beta \log \left[\frac{\pi_{\theta}(y|x)}{\pi_{\theta_{\text{ref}}}(y|x)} \cdot Z(x) \right]$$

$$r(x,y) = \beta \log \left[\frac{\pi_{\theta}(y|x)}{\pi_{\theta_{\text{ref}}}(y|x)} \right] + \beta \log Z(x)$$

$$-\log \sigma [r(x, y^w) - r(x, y^L)]$$

$$= -\log \sigma \left[\beta \log \frac{\pi_{\theta}(y^w/x)}{\pi_{\theta_{\text{ref}}}(y^w/x)} + \beta \log \cancel{z(x)} \right]$$

$$- \beta \log \frac{\pi_{\theta}(y^L/x)}{\pi_{\theta_{\text{ref}}}(y^L/x)} - \beta \log \cancel{z(x)} \Big]$$

DPO Objective

$$\tilde{\theta} = \arg \min_{\theta} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} \left[\underbrace{\text{KL}(\pi_{\theta}(\cdot|\mathbf{x}) \parallel \pi^*(\cdot|\mathbf{x}))}_{\text{KL divergence}} - \underbrace{\log Z(\mathbf{x})}_{\text{constant wrt. } \theta} \right]$$

does not affect the result of the argmin operation, and can be removed from the objective.

$$\tilde{\theta} = \arg \min_{\theta} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} \left[\text{KL}(\pi_{\theta}(\cdot|\mathbf{x}) \parallel \pi^*(\cdot|\mathbf{x})) \right]$$

Clearly, the solution to this optimization problem is given by

$$\begin{aligned} \pi_{\theta}(\mathbf{y}|\mathbf{x}) &= \pi^*(\mathbf{y}|\mathbf{x}) \\ &= \frac{\pi_{\theta_{\text{ref}}}(\mathbf{y}|\mathbf{x}) \exp\left(\frac{1}{\beta} r(\mathbf{x}, \mathbf{y})\right)}{Z(\mathbf{x})} \end{aligned}$$

DPO Objective

$$\begin{aligned}\pi_{\theta}(\mathbf{y}|\mathbf{x}) &= \pi^*(\mathbf{y}|\mathbf{x}) \\ &= \frac{\pi_{\theta_{\text{ref}}}(\mathbf{y}|\mathbf{x}) \exp\left(\frac{1}{\beta}r(\mathbf{x}, \mathbf{y})\right)}{Z(\mathbf{x})}\end{aligned}$$

Given this equation, we can express the reward $r(\mathbf{x}, \mathbf{y})$ as

$$r(\mathbf{x}, \mathbf{y}) = \beta \left(\log \frac{\pi_{\theta}(\mathbf{y}|\mathbf{x})}{\pi_{\theta_{\text{ref}}}(\mathbf{y}|\mathbf{x})} + \log Z(\mathbf{x}) \right) \quad \rightarrow$$

Interesting because we eventually obtain a representation of the reward model based on the policy

we can apply it to the Bradley-Terry model to calculate the preference probability

DPO Objective

$$r(\mathbf{x}, \mathbf{y}) = \beta \left(\log \frac{\pi_{\theta}(\mathbf{y}|\mathbf{x})}{\pi_{\theta_{\text{ref}}}(\mathbf{y}|\mathbf{x})} + \log Z(\mathbf{x}) \right)$$

$$\begin{aligned} \Pr_{\theta}(\underline{\mathbf{y}_a} \succ \underline{\mathbf{y}_b}|\mathbf{x}) &= \text{Sigmoid}(r(\mathbf{x}, \mathbf{y}_a) - r(\mathbf{x}, \mathbf{y}_b)) \\ &= \text{Sigmoid}\left(\beta \left(\log \frac{\pi_{\theta}(\mathbf{y}_a|\mathbf{x})}{\pi_{\theta_{\text{ref}}}(\mathbf{y}_a|\mathbf{x})} + \log Z(\mathbf{x}) \right) - \right. \\ &\quad \left. \beta \left(\log \frac{\pi_{\theta}(\mathbf{y}_b|\mathbf{x})}{\pi_{\theta_{\text{ref}}}(\mathbf{y}_b|\mathbf{x})} + \log Z(\mathbf{x}) \right) \right) \\ &= \text{Sigmoid}\left(\beta \log \frac{\pi_{\theta}(\mathbf{y}_a|\mathbf{x})}{\pi_{\theta_{\text{ref}}}(\mathbf{y}_a|\mathbf{x})} - \beta \log \frac{\pi_{\theta}(\mathbf{y}_b|\mathbf{x})}{\pi_{\theta_{\text{ref}}}(\mathbf{y}_b|\mathbf{x})} \right) \end{aligned}$$

DPO Objective

$$\Pr_{\theta}(\mathbf{y}_a \succ \mathbf{y}_b | \mathbf{x}) = \text{Sigmoid} \left(\beta \log \frac{\pi_{\theta}(\mathbf{y}_a | \mathbf{x})}{\pi_{\theta_{\text{ref}}}(\mathbf{y}_a | \mathbf{x})} - \beta \log \frac{\pi_{\theta}(\mathbf{y}_b | \mathbf{x})}{\pi_{\theta_{\text{ref}}}(\mathbf{y}_b | \mathbf{x})} \right)$$

This formula is elegant because it converts the difference in rewards into the difference in ratio functions, and we do not need to calculate the value of $Z(\mathbf{x})$.

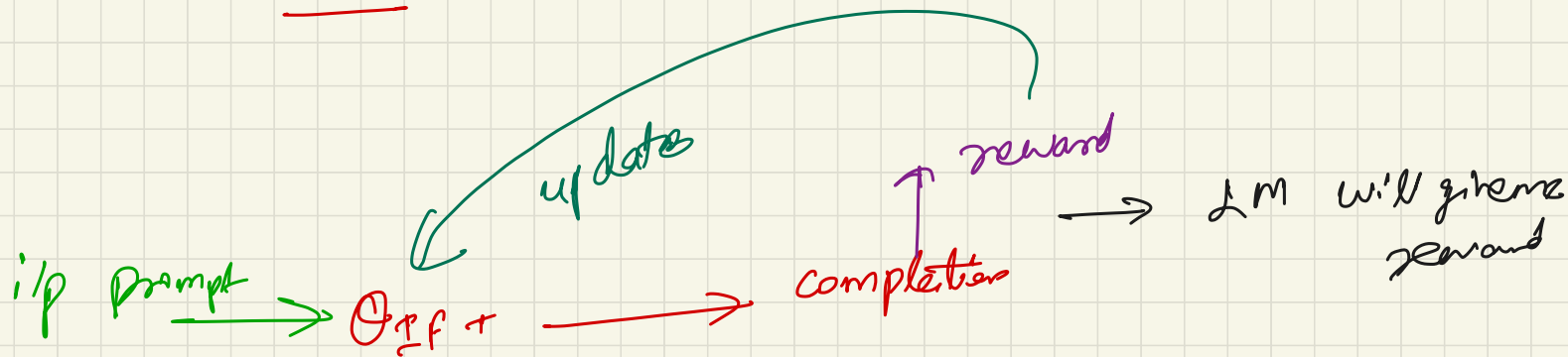
Finally, we can train the target policy by minimizing the following DPO loss function

$$\mathcal{L}_{\text{dpo}}(\theta) = -\mathbb{E}_{(\mathbf{x}, \mathbf{y}_a, \mathbf{y}_b) \sim \mathcal{D}_r} [\log \Pr_{\theta}(\mathbf{y}_a \succ \mathbf{y}_b | \mathbf{x})]$$

It should be noted that the loss function here depends on the parameters of the policy (i.e., θ) rather than the parameters of the reward model

$\Theta_{\text{rand}} \rightarrow \Theta_{\text{pr}} \rightarrow \Theta_{\text{IFP}}$

$$|w - y_e|$$



$$\Theta_{\text{new}} = \Theta + \alpha \left[\nabla E(R(s)) \right]$$

learning rate

PPO: Thinking in terms of trajectories

A trajectory is a series of (action, state), starting from an initial state

$$\tau = (s_0, a_0, s_1, a_1, \dots).$$

We will model the next state as being stochastic

$$s_{t+1} \sim P(\cdot | s_t, a_t).$$

We can thus define the probability of a trajectory as follows:

$$P(\tau | \pi) = \rho_0(s_0) \prod_{t=0}^{T-1} P(s_{t+1} | s_t, a_t) \pi(a_t | s_t).$$

We will always work with discounted rewards (we prefer immediate rewards instead of future):

$$R(\tau) = \sum_{t=0}^{\infty} \gamma^t r_t.$$